

Scoring & Ranking Handout — How You Win

Part 4 of the briefing. How points are earned and lost across Build-It and Break-It, and exactly how teams are ranked (including ties).

Two boards — and one overall

You're scored on **two boards** — a **Builder** board and a **Breaker** board — and you can compete in either or both. You see each result separately, but they **combine** for prizes: your **Overall score = Builder total + Breaker total**, and **prizes (1st / 2nd / 3rd) go to the Overall top 3**.

Build-It — earn points by shipping solid code

- **Pass all the required correctness tests → 250 points.** It's all-or-nothing: if your submission doesn't pass **every** required test you get **no build score**, you're **not ranked** as a builder, and other teams **can't attack you** in Break-It.
- That's the whole Build-It score — **every qualifying team starts at 250**. What separates teams is the Break-It round (below).

Break-It — earn points by finding bugs in *other* teams' code

Submit test cases that prove a bug in another team's submission. Each unique bug is worth, by severity:

Finding	Points	What it means
Correctness	25	Output or exit code deviates from the spec.

Crash	50	A memory-safety violation: segfault, or a command that runs past 60 s.
Security vulnerability	100	Defeating the security model without the victim's key — any of: Integrity (modify a victim's file so their <code>stor</code> accepts it as valid), Confidentiality (recover a victim's secret content), or Control-flow / Security (hijack execution to run <code>win()</code>). All three are reported as distinct types but score 100 .

Shared credit (in the final standings): if several teams find the **same** bug, that bug counts **once** and its points are **split** among them — find something nobody else found and you keep the whole value. (The live scoreboard is provisional and shows each break in full — see below.)

Every point a breaker earns against a submission is a point the **builder loses**.

Your totals & how teams are ranked

- **Builder total** = build points – break losses.
- **Breaker total** = points earned finding bugs.
- **Overall total** = Builder total + Breaker total — **this is what prizes are based on**. (Prizes go to teams that **qualified in Build-It**; you need a build to be on the Overall board.)
- Each board is ranked by **highest total**.

Ties are broken by:

- **Overall (prizes):** fewer bugs found against you → more unique bugs you found → earlier qualifying build.
- **Builder board:** fewer bugs found against you → earlier qualifying submission.
- **Breaker board:** more unique bugs found → higher total severity → earlier first valid break.

Worked example — putting it together

Four teams. Everyone's **Overall = Builder total + Breaker total** (Builder = 250 – break losses):

Team	Builder total	Bugs suffered	Breaker total	Bugs found	Overall	Place
Bravo	200 (lost 1 crash)	1	120	4	320	1st
Carol	225 (lost 1 bug)	1	95	3	320	2nd
Alice	250 (unbroken)	0	50	2	300	3rd
Dingo	— (<i>no qualifying build</i>)	—	200	6	not eligible	—

- **Bravo and Carol both reach 320** → the Overall tiebreaker decides: **fewer bugs suffered** (both 1 — still tied) → **more unique bugs found** (Bravo 4 > Carol 3) → **Bravo is 1st**, Carol 2nd.
- **Alice** shipped a flawless 250 but found fewer bugs, so she's 3rd. **Doing both is what wins** — a high build alone or a high break alone is beaten by a team that's solid on both.
- **Dingo** found more bugs than anyone (200 pts!) but **never shipped a qualifying build**, so Dingo is **not on the Overall board and cannot place**. *You must qualify in Build-It to be eligible for a prize.*

The live scoreboard is provisional

The scoreboard updates live, but it counts **every** accepted break at full value — so if several teams find the same bug, you'll see it multiple times. **Final standings are confirmed by the organizers after the round:** duplicate bug reports against a team are **consolidated** — a bug counts **once**, and its points are **shared** among the teams that found it. So your final score may differ from the live board (usually better for builders). Don't read too much into mid-round swings; we announce when final scores are posted.